COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	2026-10-20	**DATE SUBMITTED:**	2026-10-26
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ACTIVITY NO. 12 **Build a Sample Web App in a Docker Container**

1. PROCEDURE OUTPUT

Part 1: Launch the DEVASC VM



Figure 1.1 Launching the CyberOps Workstation VM

Part 2: Create a Simple Bash Script

```
#!/bin/bash

echo -n "Enter Your Name: "
read userName
echo "Your name is $userName."
```

Figure 2.1. Bash Script

```
[analyst@secOps sample-app]$ bash user-input.sh
Enter Your Name: gabriel
Your name is gabriel.
```

Figure 2.2. Bash user-input.sh

Part 3: Create a Sample Web App

- Step 1: Install Flask and open a port on the DEVASC VM firewall

```
[analyst@secOps ~]$ pip show flask
Name: Flask
Version: 2.0.3
Summary: A simple framework for building complex web applications.
Home-page: https://palletsprojects.com/p/flask
Author: Armin Ronacher
Author-email: armin.ronacher@active-4.com
License: BSD-3-Clause
Location: /home/analyst/.local/lib/python3.6/site-packages
Requires: click, itsdangerous, Jinja2, Werkzeug
Required-by:
```

Figure 3.1. Installation of flask

- Step 2 - 6

```
from flask import Flask
from flask import request

sample = Flask(__name__)

@sample.route("/")
def main():
  return "You are calling me from " + request.remote_addr + "\n"

if __name__ == "__main__":
  sample.run(host="0.0.0.0", port=8080)
```

Figure 3.2. Python Code

- Step 7: Save and run your sample web app.

```
[analyst@secOps sample-app]$ python3 sample_app.py
 * Serving Flask app 'sample_app' (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployme
nt.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on all addresses.
   WARNING: This is a development server. Do not use it in a production deployme
nt.
 * Running on http://10.0.2.15:8080/ (Press CTRL+C to quit)
```

Figure 3.3. Running the sample_app.py

- Step 8: Verify the server is running.

```
 * Running on http://10.0.2.15:8080/ (Press CTRL+C to quit)
127.0.0.1 - - [22/Oct/2026 04:19:08] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [22/Oct/2026 04:19:09] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [22/Oct/2026 04:19:09] "GET /favicon.ico HTTP/1.1" 404 -
```

Figure 3.4. Calling 127.0.0.1

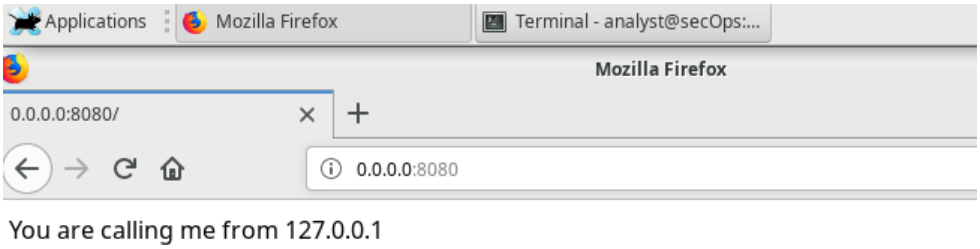


Figure 3.5. Verifying through web browser

```
[analyst@secOps ~]$ curl http://0.0.0.0:8080
You are calling me from 127.0.0.1
```

Figure 3.6. Verifying through terminal

Part 4: Configure the Web App to Use Website Files

- Step 1: Explore the directories that will be used by the web app.

```
[analyst@secOps sample-app]$ cat templates/index.html
<html>
<head>
        <title>Sample app</title>
        <link rel ="stylesheet" href="/static/style.css" />
</head>
<body>
        <h1>You are calling me from {{request.remote_addr}}</h1>
</body>
</html>
```

Figure. 4.1. index.html

- Step 2: Update the Python code for the sample web app.

```
from flask import Flask
from flask import request
from flask import render_template

sample = Flask(__name__)

@sample.route("/")
def main():
  return render_template("index.html")

if __name__ == "__main__":
  sample.run(host="0.0.0.0", port=8080)
```

Figure 4.2. Updating the sample_app.py

- Step 3: Save and run your script

```
[analyst@secOps sample-app]$ python sample_app.py
 * Serving Flask app 'sample_app' (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployme
nt.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on all addresses.
   WARNING: This is a development server. Do not use it in a production deployme
nt.
 * Running on http://10.0.2.15:8080/ (Press CTRL+C to quit)
```

Figure 4.3. Running the sample_app.py

- Step 4: Verify your program is running.

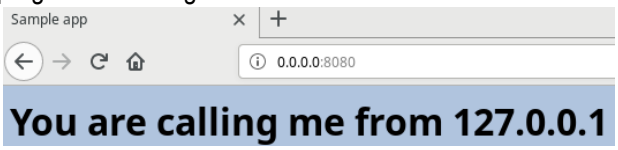


Figure 4.4. Verifying through Browser

Part 5: Create a Bash Script to Build and Run a Docker Container

```
#!/bin/bash

docker stop samplerunning 2>/dev/null
docker rm samplerunning 2>/dev/null
rm -rf tempdir

mkdir tempdir
mkdir tempdir/templates
mkdir tempdir/static

cp sample_app.py tempdir/.
cp -r templates/* tempdir/templates/.
cp -r static/* tempdir/static/.

echo "flask" > tempdir/requirements.txt

cat << 'EOF' > tempdir/Dockerfile
FROM python:3
WORKDIR /home/myapp
COPY requirements.txt .
RUN python3 -m pip install --no-cache-dir -r requirements.txt
COPY ./static ./static/
COPY ./templates ./templates/
COPY sample_app.py .
EXPOSE 8080
CMD ["python3", "sample_app.py"]
EOF

cd tempdir
docker build --no-cache -t sampleapp .
docker run -t -d -p 8080:8080 --name samplerunning sampleapp
docker ps -a
```

Figure 5.1. Bash script for sample-app.sh

Part 6: Build, Run, and Verify the Docker Container

- Step 1: Execute the bash script.

```
[analyst@secOps sample-app]$ bash ./sample-app.sh
mkdir: cannot create directory 'tempdir': File exists
mkdir: cannot create directory 'tempdir/templates': File exists
mkdir: cannot create directory 'tempdir/static': File exists
DEPRECATED: The legacy builder is deprecated and will be removed in a future rel
ease.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   7.68kB
Step 1/24 : FROM python
latest: Pulling from library/python
8d0037029946: Downloading  41.14MB/49.39MB
63040bd9752a: Download complete
7046f4e749f0: Downloading  54.62MB/67.82MB
992d9e06c2e8: Downloading  5.382MB/237MB
6f634d75d986: Waiting
23346f0ec4f3: Waiting
3c16dcb966e6: Waiting
```

Figure 6.1. Executing bash script

- Step 2: Investigate the running Docker container and the web app.

```
[analyst@secOps sample-app]$ ls tempdir/
Dockefile  Dockerfile  sample_app.py  static  templates
```

Figure 6.2. Verifying tempdir

```
[analyst@secOps sample-app]$ cat tempdir/Dockerfile
FROM python
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
FROM python
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
FROM python
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
FROM python
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
```

Figure 6.3. Verifying tempdir/Dockerfile

```
[analyst@secOps sample-app]$ bash sample-app.sh
samplerunning
samplerunning
DEPRECATED: The legacy builder is deprecated and will be removed in a future rel
ease.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  7.168kB
Step 1/9 : FROM python:3
 ---> 967555bbc8c4
Step 2/9 : WORKDIR /home/myapp
 ---> Running in af8837b3c50e
 ---> Removed intermediate container af8837b3c50e
 ---> 28b71767e3b3
Step 3/9 : COPY requirements.txt .
 ---> e1d0d456976d
Step 4/9 : RUN python3 -m pip install --no-cache-dir -r requirements.txt
 ---> Running in b86f60e56890
Collecting flask (from -r requirements.txt (line 1))
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from flask->-r requirements.txt (line 1))
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask->-r requirements.txt (line 1))
  Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask->-r requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask->-r requirements.txt (line 1))
  Downloading markupsafe-3.0.4-cp315-cp315-manylinux2014_x86_64.manylinux_2_17_x
86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Collecting werkzeug>=3.1.0 (from flask->-r requirements.txt (line 1))
  Downloading werkzeug-3.1.9-py3-none-any.whl.metadata (4.1 kB)
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.5.0-py3-none-any.whl (125 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading markupsafe-3.0.4-cp315-cp315-manylinux2014_x86_64.manylinux_2_17_x86
_64.manylinux_2_28_x86_64.whl (23 kB)
Downloading werkzeug-3.1.9-py3-none-any.whl (228 kB)
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeu
g, jinja2, flask
```

Figure 6.4. bash sample-app.sh

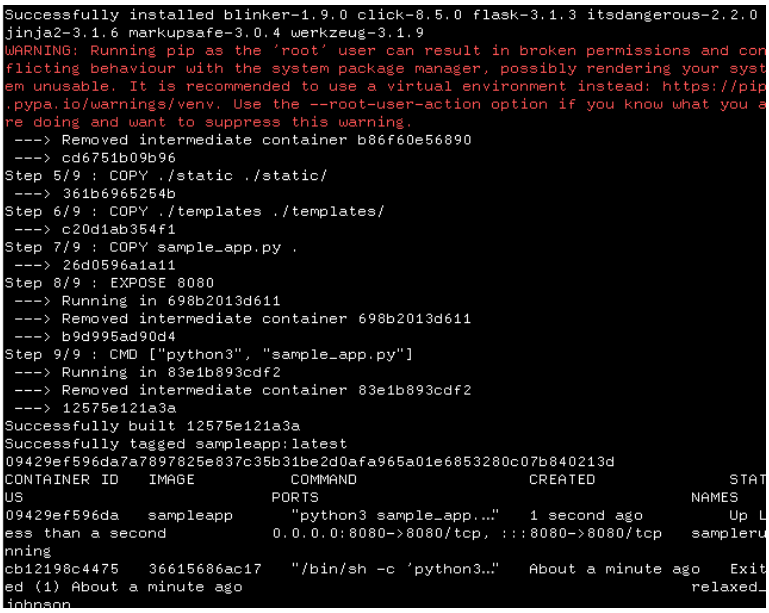


Figure 6.5. bash sample-app.sh

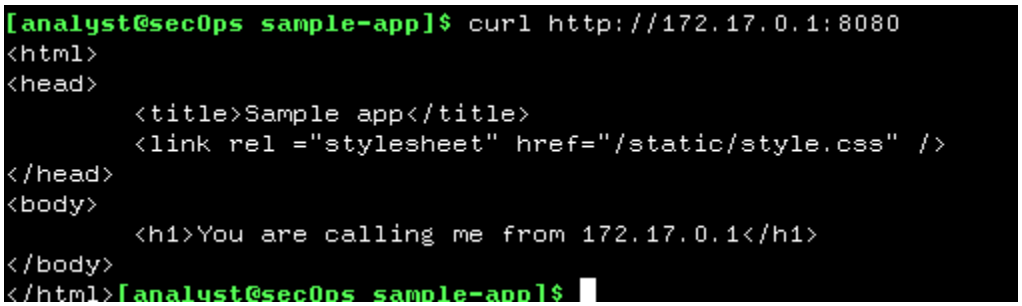


Figure 6.6. curl http://172.17.0.1:8080

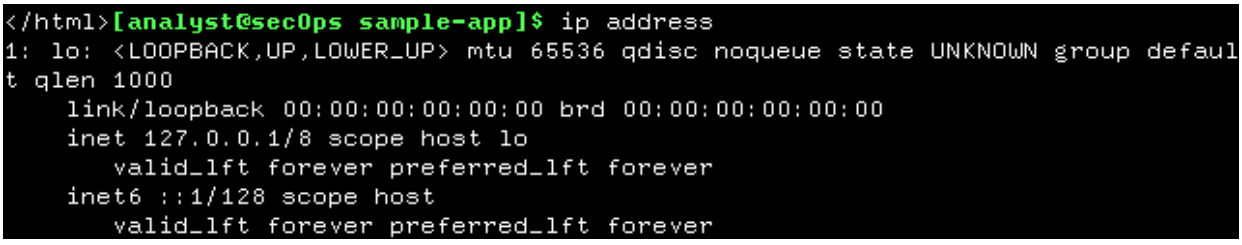


Figure 6.7. ip address

- Step 3: Access and explore the running container.

```
[analyst@secOps sample-app]$ docker exec -it samplerunning bash
root@09429ef596da:/home/myapp# pwd
/home/myapp
root@09429ef596da:/home/myapp# ls -l
total 16
-rw-r--r-- 1 root root    6 Oct 26 09:27 requirements.txt
-rw-r--r-- 1 root root  249 Oct 26 09:27 sample_app.py
drwxr-xr-x 2 root root 4096 Oct 26 09:27 static
drwxr-xr-x 2 root root 4096 Oct 26 09:27 templates
root@09429ef596da:/home/myapp# ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  3.7 120064 37764 pts/0    Ss+  09:27   0:00 python3 sample_
root           8  0.1  0.3   4316  3664 pts/1    Ss   09:36   0:00 bash
root          14  0.0  0.3   6388  3632 pts/1    R+   09:37   0:00 ps aux
root@09429ef596da:/home/myapp#
```

Figure 6.8. docker exec -it samplerunning bash

- Step 4: Stop and remove the Docker container.

```
[analyst@secOps sample-app]$ docker stop samplerunning
samplerunning
[analyst@secOps sample-app]$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS
                    PORTS     NAMES
92ba0b5e6604   sampleapp      "python3 sample_app.…"   8 minutes ago   Exited (1
37) 6 seconds ago             samplerunning
cb12198c4475   36615686ac17   "/bin/sh -c 'python3…"   6 hours ago     Exited (1
) 6 hours ago                 relaxed_johnson
```

Figure 6.9. docker stop samplerunning

```
[analyst@secOps sample-app]$ docker start samplerunning
samplerunning
```

Figure 6.10. docker start samplerunning

```
[analyst@secOps sample-app]$ docker stop samplerunning
samplerunning
[analyst@secOps sample-app]$ docker rm samplerunning
samplerunning
[analyst@secOps sample-app]$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS
              PORTS     NAMES
cb12198c4475   36615686ac17   "/bin/sh -c 'python3…"   6 hours ago   Exited (1)
6 hours ago              relaxed_johnson
```

Figure 6.11. docker rm samplerunning

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Conclusion:

- Through this lab, a customized shell script called `sample-app.sh` allowed the creation, execution and testing of a containerized Flask application. The constraints of the host system's package manager and potential Dockerfile construction pitfalls like properly dealing with JSON, quote escaping, and the `requirements.txt` file were successfully managed, and this enabled us to establish a reliable, automated deployment process. We established that the container will be continuously available to respond to HTTP `curl` requests on port 8080.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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